CSE 429

Technical Writing and Presentation

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LaTeX

Equations

- For the equation, must add a userpackage

  \usepackage{amsmath}

- To write an equation, an **{equation}** environment must needed.

```
\begin{equation}
Equation
\end{equation}
```

To unnumbered the equation simply put an asterisk (*) at the end of the word equation.

Simple math formula

x^2 `$x^2$` → **Superscripts** created by ^

x_2 `$x_2$` → **Subscripts** created by _

x^{n+2} `$x^{n+2}$` x_{n+2} `$x_{n+2}$`

x_{n+2}^{n+2} `$x_{n+2}^{n+2}$`

$\sum_{n=0}^{\infty} n$ `$ \sum_{n=0}^\infty n$`

Simple math formula

- Powers:

```
e^{i\pi} = -1
```

$$e^{i\pi} = -1$$

- Subscripts:

```
a_{n+1} = 2 \times a_n
```

$$a_{n+1} = 2 \times a_n$$

- Fractions:

```
\frac{x^\pi}{a_{25}}
```

$$\frac{x^\pi}{a_{25}}$$

- Summation:

```
\sum_{i=0}^{\infty} a^i
```

$$\sum_{i=0}^{\infty} a^i$$

Equation (self-study)

Inline math mode

You can use any of these "delimiters" to typeset your math in inline mode:

- `\(...\)`
- `$...$`
- `\begin{math}...\end{math}`

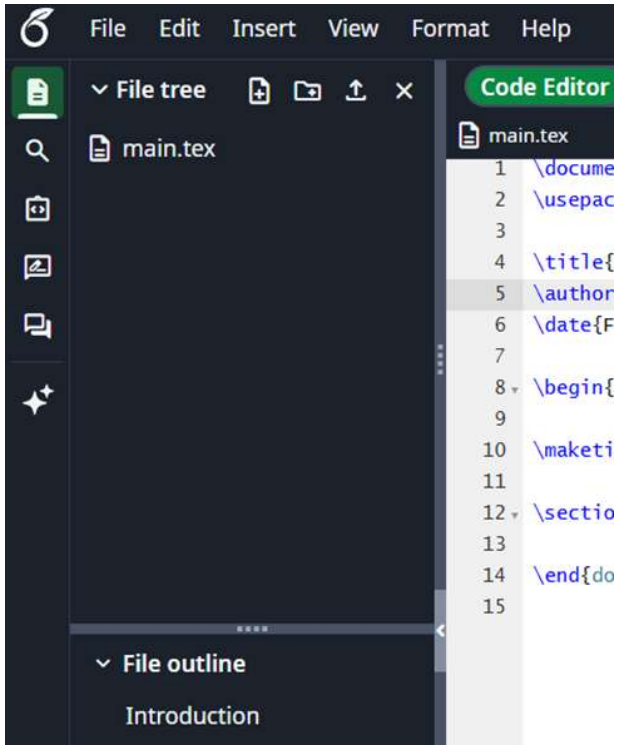
They all work and the choice is a matter of taste, so let's see some examples.

Mathematical expressions: https://www.overleaf.com/learn/latex/Mathematical_expressions

Image

- First, upload the image in the **File Tree pane**
- For inserting image the userpackage **\usepackage{graphicx}**
- For image, a **{figure}** environment must needed.

```
\begin{figure}[h]
    ..............
    ..............
\end{figure}
```



Image

- **[h]** = Place the float here, i.e., approximately at the same point it occurs in the source text
- **\centering** = alignment
- **\includegraphics** = holds the image size parameters and path
  [size]{image location}
- **\caption** = figure name, **\label** = for referencing

```
\begin{figure}[h]
    \centering
    \includegraphics[width=0.75\textwidth]{mesh}
    \caption{A nice plot.}
    \label{fig:mesh1}
\end{figure}
```

Image position

Parameter	Position
h	Place the float *here*, i.e., *approximately* at the same point it occurs in the source text (however, not *exactly* at the spot)
t	Position at the *top* of the page.
b	Position at the *bottom* of the page.
p	Put on a special *page* for floats only.
!	Override internal parameters LaTeX uses for determining "good" float positions.
H	Places the float at precisely the location in the LATEX code. Requires the float package. This is somewhat equivalent to h!.

Multiple Image (Self study)

- Multiple images in one figure
- Wrapping text around a figure

https://www.overleaf.com/learn/latex/Positioning_images_and_tables#Positioning_images

Lab Task 4 (Inserting Equations & Figures)

Write a report, use 3-4 equations (both numbered and unnumbered), include some inline equations, and add figures, including multiple figures and one figure with wrapped text.

In the lab report,
Add: Cover page, Software, and describe the commands used code.

Result: Provide a screenshot of the code and merge the PDF result part.

Challenges (you've faced regarding the task).

Thank You